

The critical second Weyl coefficient of a harmonic delta chain

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Abstract

We determine the second high-energy eigenvalue-counting coefficient for a half-line harmonic chain with positive point interactions at $x_n = \pi \sum_{j=1}^n j^{-1}$. If the couplings are fixed independently of the spectral parameter and satisfy $b_n/n \rightarrow \kappa \in (0, \infty)$, then the inclusive count obeys $N_b(k^2) = k \log k + C(\kappa)k + o(k)$ as real $k \rightarrow \infty$. The coefficient is a convergent regularized integral of the integrated density of states of a periodic comparison problem with cell length π and coupling κ . Its derivation requires control of a growing transition region in which finite delta transmission contributes at order k . We prove a uniform finite-chain counting estimate and a local-periodic reduction using geometric blocks, an explicit Rayleigh-quotient comparison, and a telescoping bound on the accumulated freezing error. The function C is continuous and strictly decreasing, with range $(2\gamma - 1, \infty)$, where γ is Euler's constant. The formula is stable under arbitrary positive $o(n)$ perturbations of the couplings. If $b_n/n \rightarrow \infty$, the second coefficient is $2\gamma - 1$. If $b_n/n \rightarrow 0$ and $nb_n \rightarrow \infty$, we prove only that $[N_b(k^2) - k \log k]/k \rightarrow \infty$; no general soft-regime leading asymptotic is asserted.

1 Introduction and main results

Point interactions on a chain of shrinking intervals can remain visible in the second term of a high-energy counting law. On the harmonic chain, the critical balance occurs when the interaction at the n th vertex has strength comparable to n . We prove that the resulting second coefficient is determined by the full periodic-cell integrated density of states, rather than by its free or Dirichlet endpoint.

The harmonic-chain model is classical. Egger né Endres and Steiner [3] considered constant positive coupling, proved discreteness, and analyzed in detail the fully decoupled Dirichlet limit. Its eigenvalues are the squares of positive integers with divisor multiplicities, so its counting law is an instance of Dirichlet's divisor problem. General realization and discreteness results for shrinking-gap point interactions are developed by Kostenko and Malamud [4]. We include a direct form argument because its quantitative tail bound is needed for the counting proof, not because compactness itself is new.

The periodic comparison uses the usual delta-jump and free-propagation matrices; their constant-coupling specialization and band structure are recalled, for example, in Drabkin, Kirsch, and Schulz-Baldes [2, Secs. 2.1–2.3]. The spectral comparison work of Bifulco and Kerner [1] also revisits the harmonic chain. Its modified local Weyl theorem in *arXiv v1, Section 5, Theorem 11* assumes fully Dirichlet coupling. That local theorem does not provide the global critical second coefficient considered here. These comparisons identify the background used below; we make no general priority claim concerning all slowly varying periodic-operator asymptotics.

1.1 The source operator and the periodic comparison

Write

$$x_0 = 0, \quad x_n = \pi H_n, \quad H_n = \sum_{j=1}^n \frac{1}{j}, \quad I_n = (x_{n-1}, x_n). \quad (1.1)$$

Thus $|I_n| = \pi/n$, while $x_n \rightarrow \infty$. Throughout the paper, $(b_n)_{n \geq 1}$ is a sequence of positive real numbers, fixed independently of the spectral frequency k , and

$$nb_n \rightarrow \infty. \quad (1.2)$$

Let

$$\begin{aligned} \mathcal{V} &= \{f \in H^1(0, \infty) : f(0) = 0\}, \\ q_b[f] &= \int_0^\infty |f'(x)|^2 dx + \sum_{n \geq 1} b_n |f(x_n)|^2, \\ \mathcal{D}_b &= \left\{ f \in \mathcal{V} : \sum_{n \geq 1} b_n |f(x_n)|^2 < \infty \right\}. \end{aligned} \quad (1.3)$$

Only the trace at zero is prescribed; there is no additional boundary condition at infinity. Section 2 proves that this form is densely defined and closed, with compact embedding into $L^2(0, \infty)$. Its represented self-adjoint operator is denoted by H_b . It is the form realization of $-d^2/dx^2 + \sum_n b_n \delta_{x_n}$, with continuity and derivative jump $f'(x_n+) - f'(x_n-) = b_n f(x_n)$ at each vertex. The associated source dynamics is $\exp(-itH_b)$; our observable is the inclusive spectral count

$$N_b(k^2) = \#\{j : \lambda_j(H_b) \leq k^2\}, \quad k \geq 0, \quad (1.4)$$

where eigenvalues are counted with multiplicity.

For $\kappa > 0$ and $\theta \in [0, 2\pi)$, define the comparison-cell form

$$\begin{aligned} \mathcal{D}_\theta &= \{u \in H^1(0, \pi) : u(\pi) = e^{i\theta} u(0)\}, \\ q_{\kappa, \theta}[u] &= \int_0^\pi |u'(x)|^2 dx + \kappa |u(0)|^2. \end{aligned} \quad (1.5)$$

Let $\nu_\kappa(z, \theta)$ count its eigenvalues at most z^2 , and put

$$J_\kappa(z) = \frac{1}{2\pi} \int_0^{2\pi} \nu_\kappa(z, \theta) d\theta, \quad z \geq 0. \quad (1.6)$$

This is the integrated density of states *per cell of length* π , not per unit length. The periodic operator is a local comparison problem, not a claimed global conjugate of H_b .

Theorem 1.1 (Critical second coefficient). *Under (1.2), H_b has strictly positive discrete spectrum. If*

$$\frac{b_n}{n} \rightarrow \kappa \in (0, \infty), \quad (1.7)$$

then, as real $k \rightarrow \infty$,

$$N_b(k^2) = k \log k + C(\kappa)k + o(k), \quad (1.8)$$

where

$$C(\kappa) = \gamma + \int_0^\infty \frac{J_\kappa(z) - z \mathbf{1}_{[1, \infty)}(z)}{z^2} dz. \quad (1.9)$$

The integral converges. The function $C : (0, \infty) \rightarrow \mathbb{R}$ is continuous and strictly decreasing, with

$$C((0, \infty)) = (2\gamma - 1, \infty), \quad \lim_{\kappa \rightarrow \infty} C(\kappa) = 2\gamma - 1, \quad \lim_{\kappa \downarrow 0} C(\kappa) = \infty. \quad (1.10)$$

No rate or monotonicity is required in (1.7). In particular, the conclusion holds for every positive sequence $b_n = \kappa n + o(n)$.

Theorem 1.2 (Hard and soft limits). *Retain positivity of (b_n) and (1.2). If $b_n/n \rightarrow \infty$, then*

$$N_b(k^2) = k \log k + (2\gamma - 1)k + o(k). \quad (1.11)$$

If $b_n/n \rightarrow 0$, then the conclusion is precisely

$$\frac{N_b(k^2) - k \log k}{k} \rightarrow \infty. \quad (1.12)$$

All limits include eigenvalue thresholds. In the second case no general leading-order formula for $N_b(k^2)$ is asserted.

Fixed positive couplings $b_n \equiv b$ belong to the soft side, not to (1.7). Passing to a finite nonzero limit of b_n/n changes the asymptotic balance: for indices n of order k , the product $b_n|I_n|$ approaches $\pi\kappa$. Finite transmission therefore persists on an order- k set of cells. An endpoint replacement on those cells can change the coefficient of k even when it leaves the leading term unchanged.

1.2 The counting mechanism

The principal analytic assertion is

$$N_{(\kappa n)}(k^2) = \sum_{n \geq 1} J_\kappa(k/n) + o(k). \quad (1.13)$$

Here $N_{(\kappa n)}$ denotes the count for $b_n = \kappa n$ exactly. A bounded error on each individual cell would accumulate to order k , too large for (1.8). We instead cut into geometric blocks, use a uniform finite-chain Floquet estimate, and control the total variation between block endpoints by summation by parts. This proves (1.13) without assuming that the periodic density of states is smooth at band edges.

Section 2 establishes the source form and cut comparison. Section 3 supplies the periodic estimates, including the uniform finite-chain error. The reduction is proved in Section 4; Section 5 evaluates and analyzes its second coefficient. Section 6 proves the asymptotic-coupling and endpoint statements by form comparison, with the domain inclusions made explicit.

2 The closed form, compactness, and cuts

Proposition 2.1 (Quantitative compactness). *The form (1.3) is densely defined, closed, and nonnegative. Under (1.2), its domain embeds compactly into $L^2(0, \infty)$. For every integer $M \geq 1$,*

$$\int_{x_M}^{\infty} |f(x)|^2 dx \leq a_M q_b[f], \quad a_M = 2\pi \sup_{n > M} \frac{1}{nb_n} + \frac{2\pi^2}{(M+1)^2}, \quad (2.1)$$

for $f \in \mathcal{D}_b$, and $a_M \rightarrow 0$. The represented operator has a positive first eigenvalue and discrete spectrum tending to infinity.

Proof. Since $x_n \rightarrow \infty$, a compact interval meets only finitely many vertices. Thus $C_c^\infty(0, \infty) \subset \mathcal{D}_b$, proving density. Suppose (f_j) is Cauchy for the norm $(\|f\|_2^2 + q_b[f])^{1/2}$. It converges to some $f \in \mathcal{V}$ in $H^1(0, \infty)$, and $(\sqrt{b_n}f_j(x_n))_{n \geq 1}$ converges in ℓ^2 . For each fixed n , point evaluation is continuous on a finite interval containing x_n . Therefore the n th coordinate of the second limit is $\sqrt{b_n}f(x_n)$. Hence $f \in \mathcal{D}_b$ and f_j converges in the form norm. This proves closedness.

For $x \in I_n$, the fundamental theorem for Sobolev functions and Cauchy–Schwarz give

$$|f(x)|^2 \leq 2|f(x_n)|^2 + 2|I_n| \int_{I_n} |f'(y)|^2 dy.$$

After integration,

$$\int_{I_n} |f(x)|^2 dx \leq \frac{2\pi}{n} |f(x_n)|^2 + \frac{2\pi^2}{n^2} \int_{I_n} |f'(x)|^2 dx. \quad (2.2)$$

Summing over $n > M$ proves (2.1). Assumption (1.2) makes $a_M \rightarrow 0$. The form unit ball is bounded in $H^1(0, x_M)$ for each M , so Rellich compactness gives local L^2 compactness. The uniform tail bound upgrades it to compactness in $L^2(0, \infty)$.

The compact form embedding gives compact resolvent. If $q_b[f] = 0$, then $f' = 0$ almost everywhere, and the zero left trace forces $f = 0$. Thus zero is not an eigenvalue. The compact spectral theorem now gives the final assertions. $\square$

Lemma 2.2 (Finite-codimension count comparison). *Impose zero traces at r specified internal vertices of the chain. Let $N_b^D(E)$ be the inclusive eigenvalue count of the restricted form. For every real E ,*

$$N_b(E) - r \leq N_b^D(E) \leq N_b(E). \quad (2.3)$$

The restricted form is the orthogonal sum of the Dirichlet-ended block forms and, if present, the remaining tail form.

Proof. Each point trace is a bounded linear functional on the form domain. The joint kernel therefore has codimension at most r . If V_E is the span of the original eigenvectors of eigenvalues at most E , its intersection with that kernel has dimension at least $\dim V_E - r$. The Rayleigh quotient on this intersection is at most E . Conversely, every subspace of the restricted form domain is a subspace of the original one. The variational characterization of the inclusive count proves (2.3), including at eigenvalue thresholds. At a cut the corresponding delta term vanishes; continuity and the zero traces identify the restricted form with the stated direct sum. $\square$

Remark 2.3. An infinite set of cuts is not a finite-codimension perturbation. When the fully Dirichlet operator is used below, its comparison with H_b is by form-domain inclusion, not by a finite count error.

3 The periodic cell and a uniform finite-chain estimate

The cell forms (1.5) are closed on their stated domains, have compact resolvent, and depend continuously on their parameters. We first record bounds with the normalization needed for the infinite-chain calculation.

Lemma 3.1 (Cell bounds and band bottom). *For every $\kappa > 0$, $\theta \in [0, 2\pi)$, and $z \geq 0$,*

$$\lfloor z \rfloor \leq \nu_\kappa(z, \theta) \leq \lfloor z \rfloor + 1, \quad \lfloor z \rfloor \leq J_\kappa(z) \leq z. \quad (3.1)$$

There is a unique $r(\kappa) \in (0, 1)$ satisfying

$$r(\kappa) \tan \frac{\pi r(\kappa)}{2} = \frac{\kappa}{2}. \quad (3.2)$$

It is continuous and strictly increasing in κ , and

$$J_\kappa(z) = 0 \quad (0 \leq z < r(\kappa)), \quad J_\kappa(z) > 0 \quad (z > r(\kappa)). \quad (3.3)$$

The function J_κ is nondecreasing in z and nonincreasing in κ .

Proof. The Dirichlet domain $H_0^1(0, \pi)$ is the kernel of the single trace $u \mapsto u(0)$ on $\mathcal{D}_\theta$. It is therefore a codimension-one restriction, with eigenvalues j^2 , $j \geq 1$. The same variational argument as in Lemma 2.2 proves the first two-sided bound in (3.1). Increasing κ increases the form on its fixed domain, so all fiber eigenvalues are nondecreasing in κ . At $\kappa = 0$, the eigenfrequencies are $|2j + \theta/\pi|$, $j \in \mathbb{Z}$. Tonelli's theorem gives

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} \sum_{j \in \mathbb{Z}} \mathbf{1}_{\{|2j + \theta/\pi| \leq z\}} d\theta &= \frac{1}{2} \sum_{j \in \mathbb{Z}} \int_0^2 \mathbf{1}_{\{|2j + s| \leq z\}} ds \\ &= \frac{1}{2} \int_{\mathbb{R}} \mathbf{1}_{[-z, z]}(s) ds = z. \end{aligned}$$

Comparison with these free forms and integration of the Dirichlet lower bound prove the remaining inequalities in (3.1). Monotonicity in energy is immediate from the definition of the count.

The endpoint estimate on a cell gives, uniformly in θ ,

$$\|u\|_2^2 \leq 2\pi|u(0)|^2 + 2\pi^2\|u'\|_2^2 \leq \left(\frac{2\pi}{\kappa} + 2\pi^2\right) q_{\kappa, \theta}[u]. \quad (3.4)$$

There is consequently a positive lower bound for all fiber eigenvalues. A quasiperiodic function u has periodic modulus $|u|$, with the same L^2 norm and endpoint penalty and with no larger derivative energy. Thus the bottom over all fibers is the periodic ground eigenvalue.

In the periodic fiber, the positive symmetric solution of the interior eigenvalue equation is a multiple of $\cos(z(x - \pi/2))$. Its endpoint derivative condition is

$$u'(0) - u'(\pi) = \kappa u(0),$$

which gives $2z \sin(\pi z/2) = \kappa \cos(\pi z/2)$. The function $z \mapsto z \tan(\pi z/2)$ increases continuously from zero to infinity on $(0, 1)$, so it has the unique solution (3.2). The corresponding eigenfunction is strictly positive, and hence is the ground state by the one-dimensional variational/Sturm characterization. This proves the vanishing in (3.3) and the asserted dependence of r on κ .

For completeness, continuity of the fiber eigenvalues in θ follows on writing $u(x) = e^{i\theta x/\pi} v(x)$ with v periodic. The form becomes

$$\int_0^\pi |v'(x) + i\theta v(x)/\pi|^2 dx + \kappa |v(0)|^2$$

on a fixed domain. If $z > r(\kappa)$, the periodic ground eigenvalue is strictly below z^2 , so the same is true on an open set of nearby phases. The phase integral is therefore positive, proving the second statement in (3.3). $\square$

Lemma 3.2 (Phase equation and continuity of the cell count). *At a fixed positive frequency z , the phases for which z^2 is a fiber eigenvalue satisfy*

$$D_\kappa(z) := \cos(\pi z) + \frac{\kappa}{2z} \sin(\pi z) = \cos \theta. \quad (3.5)$$

There are at most two such phases modulo 2π . The integrated count $J_\kappa(z)$ is continuous in $z \geq 0$.

Proof. For the state vector $(u, u')^\top$, propagation across one interval and passage through its terminal delta interaction are represented by

$$P_z = \begin{pmatrix} \cos(\pi z) & z^{-1} \sin(\pi z) \\ -z \sin(\pi z) & \cos(\pi z) \end{pmatrix}, \quad K_\kappa = \begin{pmatrix} 1 & 0 \\ \kappa & 1 \end{pmatrix}. \quad (3.6)$$

Both matrices have determinant one. The quasiperiodic value and derivative conditions obtained from (1.5) state that $e^{i\theta}$ is an eigenvalue of $K_\kappa P_z$. Taking the trace proves (3.5).

At an integer $z = m \geq 1$, $P_z = (-1)^m I$, but $K_\kappa \neq I$. Thus the transfer matrix is a nontrivial shear with the sole eigenphase 0 or π , and its eigenspace has dimension one. No additional phase or double free eigenspace is lost at those frequencies. At zero, the continuous matrix limit has half-trace $1 + \kappa\pi/2 > 1$, consistently with (3.4).

Fix $z_0 > 0$. Except for the at most two phases in (3.5) at z_0 , the fiber count is constant for z sufficiently close to z_0 . On a bounded interval of frequencies the first bound in (3.1) provides a common integrable bound on the phase counts. Dominated convergence proves continuity of J_κ at z_0 ; the uniform gap gives continuity at zero. $\square$

Proposition 3.3 (Uniform finite-chain counting error). *Let $N_{M,\kappa}^D(z^2)$ be the inclusive count on a chain of M cells of length π , with coupling κ at every internal junction and Dirichlet conditions at both ends. For all integers $M \geq 1$, all $\kappa > 0$, and all $z \geq 0$,*

$$|N_{M,\kappa}^D(z^2) - MJ_\kappa(z)| \leq 5. \quad (3.7)$$

Proof. At fixed $z > 0$, the fiber count is locally constant in θ away from the phases in (3.5). This follows from continuity of the eigenvalues after the fixed-domain gauge used above. By (3.1), the difference

$$h(\theta) = \nu_\kappa(z, \theta) - \lfloor z \rfloor$$

takes only the values zero and one. With at most two exceptional phases, it is the indicator of a circle arc or its complement, or a constant function, up to values at the exceptional points. The number of points of a uniform M -point phase mesh in a circle arc differs from M times its normalized length by at most two. Altering values at at most two endpoints changes the sum by at most two more. Hence

$$\left| \sum_{j=0}^{M-1} \nu_\kappa(z, 2\pi j/M) - MJ_\kappa(z) \right| \leq 4. \quad (3.8)$$

The loose constant covers inclusive endpoint counts as well.

Join the ends of an M -cell chain into a ring, with one delta interaction of strength κ at each of its M junctions. Translation by one cell is unitary, has order M , and commutes with the form. Its finite Fourier decomposition gives exactly the fibers at phases $2\pi j/M$. On each sector the norm and energy are both M times their one-cell counterparts, so neither the eigenvalue nor the coupling is rescaled. The sum in (3.8) is therefore the ring count.

Imposing zero trace at one specified ring junction has codimension one and cuts the ring to the Dirichlet-ended chain. The delta term at that junction vanishes, and the restricted form is exactly the chain form. Its count lies between the ring count minus one and the ring count. Combining this fact with (3.8) gives (3.7). At $z = 0$, all counts vanish by the positive gap, so the same estimate holds. $\square$

The periodic ingredients above are included to fix normalization and threshold conventions. The essential property for the nonperiodic problem is that the error in (3.7) is independent of the number of cells and of the test frequency.

4 The critical local-periodic reduction

In this section $b_n = \kappa n$ exactly, for a fixed $\kappa > 0$. We write $N_{(\kappa n)}$ for its count.

Proposition 4.1 (Local-periodic reduction). *As real $k \rightarrow \infty$,*

$$N_{(\kappa n)}(k^2) = \sum_{n \geq 1} J_\kappa(k/n) + o(k). \quad (4.1)$$

The sum has only finitely many nonzero terms for each k .

Proof. We separate the high-frequency head from the transition region and first exclude a sufficiently distant tail.

Tail exclusion. For the exact linear couplings, (2.1) gives

$$\int_{x_M}^{\infty} |f(x)|^2 dx \leq \frac{A_\kappa}{(M+1)^2} q_{(\kappa n)}[f], \quad A_\kappa = \frac{2\pi}{\kappa} + 2\pi^2. \quad (4.2)$$

Choose a fixed R satisfying

$$R^2 > A_\kappa, \quad R > r(\kappa)^{-1} + 1, \quad M = \lceil Rk \rceil. \quad (4.3)$$

On the tail with Dirichlet condition at x_M , the Rayleigh quotient is at least $(M+1)^2/A_\kappa > k^2$. One may apply (4.2) there by extending a tail function by zero to the left. Thus the tail contributes no eigenvalue at most k^2 . By Lemma 2.2, cutting at x_M changes the count by at most one.

The head. Set

$$m = \lfloor \sqrt{k} \rfloor, \quad \varepsilon = (\log k)^{-2}, \quad (4.4)$$

for k sufficiently large. The first m cells, with Dirichlet conditions at 0 and x_m , have total length πH_m . Both their coupled form and the free interval form restrict to the same sum of m Dirichlet-cell forms on imposing zero traces at the $m-1$ internal vertices. Each count is within $m-1$ of that common restricted count. The free interval count is $\lfloor k H_m \rfloor$, so

$$N_{\text{head}}(k^2) = k H_m + O(m). \quad (4.5)$$

No estimate on individual head couplings is required.

Geometric blocks. Starting with $a_0 = m$, recursively set

$$a_{i+1} = \min\{M, \lfloor (1 + \varepsilon)a_i \rfloor\}$$

until $a_B = M$. For large k , $\varepsilon m > 2$. Every block is nonempty, and, except possibly for the last shortened block, the endpoints grow by at least a factor $1 + \varepsilon/2$. In every block $a < n \leq b$,

$$0 < b - a \leq \varepsilon a, \quad \rho = \frac{b}{a} \leq 1 + \varepsilon, \quad B = O_\kappa(\varepsilon^{-1} \log k). \quad (4.6)$$

Cutting at the block boundaries and at the head endpoint costs $O(B+1)$ in the count.

Block form comparison. Fix one block $a < n \leq b$ with Dirichlet ends. Map each original cell of length π/n affinely onto a reference cell of length π/a , preserving its endpoint values. On the resulting common function space put $s_n = a/n \in [\rho^{-1}, 1]$. If v denotes the reference function, then the original norm and derivative energy are

$$\sum_{n=a+1}^b s_n \int_{\text{cell } n} |v|^2, \quad \sum_{n=a+1}^b s_n^{-1} \int_{\text{cell } n} |v'|^2,$$

respectively. Consequently

$$\rho^{-1}\|v\|_{\text{ref}}^2 \leq \|v\|_{\text{orig}}^2 \leq \|v\|_{\text{ref}}^2, \quad q_{\text{ref}}[v] \leq q_{\text{orig}}[v] \leq \rho q_{\text{ref}}[v], \quad (4.7)$$

where the reference form has coupling κa at every internal junction. Indeed, the original internal coupling κn lies between κa and $\rho \kappa a$, and the endpoint terms vanish. These affine maps give an isomorphism of the finite block form domains, so (4.7) implies the Rayleigh inequalities

$$\mathcal{R}_{\text{ref}}[v] \leq \mathcal{R}_{\text{orig}}[v] \leq \rho^2 \mathcal{R}_{\text{ref}}[v]. \quad (4.8)$$

Scaling the reference coordinate by a makes every cell have length π . Its squared L^2 norm gains factor a^{-1} and its energy factor a , while the dimensionless delta strength becomes κ . Hence its eigenvalues are a^2 times those of the $(b-a)$ -cell comparison chain. Min-max and Proposition 3.3 yield

$$(b-a)J_{\kappa}(k/b) - 5 \leq N_{\text{block}}(k^2) \leq (b-a)J_{\kappa}(k/a) + 5. \quad (4.9)$$

Accumulated freezing error. Monotonicity of J_{κ} places $\sum_{a < n \leq b} J_{\kappa}(k/n)$ between the two main terms of (4.9). Thus the total error in replacing the block counts by that discrete sum is at most $5B$ plus

$$\mathcal{E} = \sum_{i=0}^{B-1} (a_{i+1} - a_i)(j_i - j_{i+1}), \quad j_i = J_{\kappa}(k/a_i).$$

The sequence (j_i) is nonincreasing. Summation by parts, (4.6), and (3.1) give

$$\begin{aligned} \mathcal{E} &\leq \varepsilon \sum_{i=0}^{B-1} a_i(j_i - j_{i+1}) \\ &= \varepsilon \left[m j_0 + \sum_{i=1}^{B-1} (a_i - a_{i-1}) j_i - a_{B-1} j_B \right] \\ &\leq \varepsilon k \left[1 + \sum_{i=1}^{B-1} \frac{a_i - a_{i-1}}{a_i} \right] \leq \varepsilon k \left[1 + \log \frac{M}{m} \right]. \end{aligned} \quad (4.10)$$

The final inequality uses $(a_i - a_{i-1})/a_i \leq \log(a_i/a_{i-1})$. The discarded terminal term is nonpositive. In particular, no derivative or Lipschitz estimate for the IDS has been used.

Reassembling the chain. The head estimate, the finite cuts, and (4.10) imply

$$N_{(\kappa n)}(k^2) = kH_m + \sum_{m < n \leq M} J_{\kappa}(k/n) + O_{\kappa}(m + B + \varepsilon k \log k). \quad (4.11)$$

Here $m = O(\sqrt{k})$, $B = O_{\kappa}(\log^3 k)$, and $\varepsilon k \log k = k/\log k$, so the error is $o(k)$. Since $0 \leq z - J_{\kappa}(z) \leq 1$,

$$|kH_m - \sum_{n \leq m} J_{\kappa}(k/n)| \leq m.$$

For $n > M$, the cutoff in (4.3) ensures $k/n < 1/R < r(\kappa)$, and the remaining IDS terms vanish. Equation (4.11) therefore proves (4.1). $\square$

The finite chain remains periodic only inside each comparison block. The reduction neither averages the delta interactions into an ordinary scalar potential nor identifies the original operator with a globally periodic one. It retains their finite transmission precisely in the critical transition region.

5 Evaluation and dependence of the second coefficient

5.1 The regularized Riemann sum

Proposition 5.1 (The exact critical sequence). *For each fixed $\kappa > 0$, the integral in (1.9) converges and*

$$\sum_{n \geq 1} J_\kappa(k/n) = k \log k + C(\kappa)k + o(k). \quad (5.1)$$

Consequently (1.8) holds for $b_n = \kappa n$.

Proof. For $x > 0$, define

$$G_\kappa(x) = J_\kappa(1/x) - x^{-1} \mathbf{1}_{(0,1]}(x). \quad (5.2)$$

By (3.1), $|G_\kappa(x)| \leq 1$ for $0 < x \leq 1$. By (3.3), it vanishes for $x > 1/r(\kappa)$. Lemma 3.2 shows that it is continuous except possibly at the cutoff $x = 1$, away from the endpoint zero. It is thus absolutely integrable.

Its normalized Riemann sums obey

$$\frac{1}{k} \sum_{n \geq 1} G_\kappa(n/k) \longrightarrow \int_0^\infty G_\kappa(x) dx. \quad (5.3)$$

Indeed, for $0 < \delta < 1$, the integral over $(0, \delta)$ has absolute value at most δ , and the portion of the normalized mesh sum with $n/k < \delta$ is bounded by $\delta + 1/k$. On the remaining compact interval ordinary Riemann-sum convergence applies. First let $k \rightarrow \infty$ and then let $\delta \downarrow 0$. This argument does not require a limit of G_κ at zero.

For every real $k \geq 1$ there is the exact identity

$$\sum_{n \geq 1} J_\kappa(k/n) = k H_{[k]} + \sum_{n \geq 1} G_\kappa(n/k). \quad (5.4)$$

Since $H_{[k]} = \log k + \gamma + o(1)$, (5.3) proves the expansion with coefficient $\gamma + \int_0^\infty G_\kappa(x) dx$. The substitution $z = 1/x$ gives

$$\int_0^\infty G_\kappa(x) dx = \int_0^\infty \frac{J_\kappa(z) - z \mathbf{1}_{[1,\infty)}(z)}{z^2} dz.$$

Convergence also follows directly: the integrand is zero below the positive band bottom and is bounded in absolute value by z^{-2} for $z \geq 1$. This proves (5.1) and (1.9). Combining it with Proposition 4.1 proves the last assertion. $\square$

5.2 Continuity, strict monotonicity, and limits

Proposition 5.2 (Range of the coefficient). *The function C has all the properties in (1.10).*

Proof. For a fixed phase, the finite-cell forms have a fixed domain as κ varies. Point evaluation is bounded on H^1 , so their eigenvalues vary continuously at every finite $\kappa \geq 0$. At a fixed $\kappa > 0$ and $z > 0$, only finitely many phases have an eigenvalue exactly at z^2 , by Lemma 3.2. The phase counts therefore converge almost everywhere when κ varies to that value, and (3.1) dominates them. Hence $J_\kappa(z)$ is continuous in $\kappa > 0$.

On a compact positive interval of couplings, the increasing band bottom $r(\kappa)$ gives a uniform gap at zero. The z^{-2} bound at infinity is also uniform. Dominated convergence in (1.9) proves continuity of C .

If $0 < \kappa_1 < \kappa_2$, form ordering gives $J_{\kappa_1}(z) \geq J_{\kappa_2}(z)$ for all z . Moreover, $r(\kappa_1) < r(\kappa_2)$. On the nonempty interval between these band bottoms, the second integrated count is zero and the first is positive by (3.3). Therefore

$$C(\kappa_1) - C(\kappa_2) = \int_0^\infty \frac{J_{\kappa_1}(z) - J_{\kappa_2}(z)}{z^2} dz > 0. \quad (5.5)$$

As $\kappa \rightarrow \infty$, the increasing cell forms impose $u(0) = 0$ in the limit, and quasiperiodicity then imposes $u(\pi) = 0$. Their limiting form domain is the dense Dirichlet domain. Each fixed indexed eigenvalue increases to the corresponding Dirichlet eigenvalue j^2 . This also follows directly by min–max: Dirichlet test spaces give the upper bound j^2 ; any sequence of normalized eigenvectors with bounded energy is compact in L^2 and has endpoint values tending to zero, so the limiting variational spaces lie in the Dirichlet domain. Thus, for noninteger z ,

$$J_\kappa(z) \longrightarrow \lfloor z \rfloor \quad (\kappa \rightarrow \infty). \quad (5.6)$$

The uniform gap for κ bounded away from zero and the integrable tail bound justify dominated convergence in the coefficient integral. Its limiting value is

$$\int_1^\infty \frac{\lfloor z \rfloor - z}{z^2} dz = \sum_{j=1}^\infty \left[\frac{1}{j+1} - \log \frac{j+1}{j} \right] = \gamma - 1. \quad (5.7)$$

The last equality follows by taking partial sums, $H_{L+1} - 1 - \log(L+1)$, and letting $L \rightarrow \infty$. Consequently $\lim_{\kappa \rightarrow \infty} C(\kappa) = 2\gamma - 1$.

As $\kappa \downarrow 0$, finite-cell form convergence gives the free spectrum. At a fixed $z > 0$, a free fiber has eigenvalue z^2 at only finitely many phases, so the same dominated-convergence argument and the free normalization in Lemma 3.1 imply

$$J_\kappa(z) \longrightarrow z \quad (\kappa \downarrow 0). \quad (5.8)$$

The portion of the coefficient integral over $z \geq 1$ is at least -1 , by (3.1). On $0 < z < 1$ the integrand is nonnegative. For any fixed $0 < \delta < 1$, (5.8) gives

$$\lim_{\kappa \downarrow 0} \int_\delta^1 \frac{J_\kappa(z)}{z^2} dz = \int_\delta^1 \frac{dz}{z} = \log(1/\delta).$$

It follows that $\liminf_{\kappa \downarrow 0} C(\kappa) \geq \gamma - 1 + \log(1/\delta)$ for every such δ . Letting $\delta \downarrow 0$ proves the infinite endpoint limit. Strict monotonicity and continuity now give the exact range in (1.10). $\square$

6 Asymptotic couplings and the hard and soft sides

The exact critical sequence determines the full asymptotic class by min–max comparison. The order of the limiting operations is important: the spectral frequency tends to infinity before the comparison couplings approach their limiting values.

6.1 Rate-free stability at a finite positive ratio

Completion of Theorem 1.1. The form and spectral assertions follow from Proposition 2.1, and Proposition 5.2 proves the assertions about C . It remains to pass from exact linear couplings to (1.7).

Fix $0 < \eta < \kappa$. Outside a finite set F_η of vertices, the couplings satisfy

$$(\kappa - \eta)n \leq b_n \leq (\kappa + \eta)n. \quad (6.1)$$

Eventual two-sided comparability implies equality of the three form domains: weighted trace summability is equivalent on the tail, and each of the finitely many initial point penalties is finite for an H^1 function. On their common subdomain with $f(x_n) = 0$ for $n \in F_\eta$, the forms are ordered as in (6.1).

Put $r_\eta = |F_\eta|$. Combining this form ordering with Lemma 2.2 yields, for every k ,

$$N_{((\kappa+\eta)n)}(k^2) - r_\eta \leq N_b(k^2) \leq N_{((\kappa-\eta)n)}(k^2) + r_\eta. \quad (6.2)$$

Subtract $k \log k$, divide by k , and let $k \rightarrow \infty$. Proposition 5.1 gives

$$\begin{aligned} C(\kappa + \eta) &\leq \liminf_{k \rightarrow \infty} \frac{N_b(k^2) - k \log k}{k} \\ &\leq \limsup_{k \rightarrow \infty} \frac{N_b(k^2) - k \log k}{k} \leq C(\kappa - \eta). \end{aligned}$$

Continuity of C permits $\eta \downarrow 0$ and proves (1.8). The integer r_η is fixed as $k \rightarrow \infty$; no bound on its growth as $\eta \downarrow 0$ is required. This explains why the theorem imposes no rate or monotonicity in (1.7). $\square$

6.2 The fully Dirichlet comparison

Define

$$\mathcal{D}_\infty = \{f \in \mathcal{V} : f(x_n) = 0 \text{ for every } n \geq 1\}, \quad q_\infty[f] = \int_0^\infty |f'|^2 dx.$$

This is the orthogonal sum of the Dirichlet forms on the intervals I_n . Its domain is contained in every $\mathcal{D}_b$, and the forms agree there. Thus its eigenvalues lie above those of H_b in min-max order, and

$$N_D(k^2) \leq N_b(k^2). \quad (6.3)$$

This comparison uses an infinite domain restriction but does not assign it a finite-dimensional count error.

The cell eigenvalues are $(jn)^2$, $j, n \geq 1$, so the Dirichlet count is

$$N_D(k^2) = \sum_{n \leq k} \left\lfloor \frac{k}{n} \right\rfloor = k \log k + (2\gamma - 1)k + O(\sqrt{k}). \quad (6.4)$$

This is the classical divisor law underlying the model of [3]. To verify the stated error and its real- k convention directly, put $h = \lfloor \sqrt{k} \rfloor$. Counting lattice points under the hyperbola in the two coordinate strips gives

$$\sum_{n \leq k} \left\lfloor \frac{k}{n} \right\rfloor = 2 \sum_{n \leq h} \left\lfloor \frac{k}{n} \right\rfloor - h^2 = 2kH_h - h^2 + O(h).$$

The relation $H_h = \log h + \gamma + O(h^{-1})$, together with $h^2 = k + O(\sqrt{k})$, gives (6.4) for every real $k \rightarrow \infty$.

6.3 Hard couplings

Proof of (1.11). Suppose $b_n/n \rightarrow \infty$, and fix $K > 0$. Outside a finite set F_K , $b_n \geq Kn$. Here the domains need not be equal: one has

$$\mathcal{D}_b \subset \mathcal{D}_{(Kn)}. \quad (6.5)$$

Indeed, tail summability with the larger weights implies summability with the smaller ones, and finitely many initial traces are harmless. On imposing zero traces at F_K , the form q_b dominates $q_{(Kn)}$ on the smaller domain (6.5). Min-max and the finite-cut comparison therefore give

$$N_b(k^2) \leq N_{(Kn)}(k^2) + |F_K|. \quad (6.6)$$

Use (6.3) for the lower bound and Proposition 5.1 for the upper bound. After subtracting $k \log k$, dividing by k , and taking $k \rightarrow \infty$, one obtains

$$2\gamma - 1 \leq \liminf \frac{N_b(k^2) - k \log k}{k} \leq \limsup \frac{N_b(k^2) - k \log k}{k} \leq C(K).$$

Finally $K \rightarrow \infty$ and Proposition 5.2 prove (1.11). $\square$

6.4 Soft couplings

Proof of (1.12). Suppose $b_n/n \rightarrow 0$, retaining (1.2). For any fixed $K > 0$, outside a finite set F_K one has $b_n \leq Kn$. The domain inclusion is now

$$\mathcal{D}_{(Kn)} \subset \mathcal{D}_b. \quad (6.7)$$

On the domains with zero traces at F_K , the b -form is bounded above by the Kn -form on the smaller domain. Thus the variational comparison has the reverse direction from (6.6):

$$N_b(k^2) \geq N_{(Kn)}(k^2) - |F_K|. \quad (6.8)$$

It follows that

$$\liminf_{k \rightarrow \infty} \frac{N_b(k^2) - k \log k}{k} \geq C(K)$$

for every $K > 0$. Since $C(K) \rightarrow \infty$ as $K \downarrow 0$, this is precisely convergence to positive infinity in (1.12). Compactness continues to depend on the separate condition $nb_n \rightarrow \infty$. In particular this argument gives neither a discrete spectrum for the zero-coupling half-line nor a universal soft-regime leading counting law. $\square$

7 Scope and reproducibility

The critical two-term law concerns a fixed source quadratic form and its eigenvalue count. Its main input beyond the classical comparison machinery is the local-periodic reduction with cumulative $o(k)$ error. The coefficient's range, its rate-free stability, and the hard and soft statements are consequences within the same argument. The theorem does not cover arbitrary cell geometries or give a sharp oscillatory remainder.

The harmonic-chain Dirichlet benchmark has an exact divisor interpretation, but no corresponding target arithmetic identification is obtained for the coupled operator. In particular, a two-term mean counting law does not establish target Euler factors, root numbers, individual Riemann-zero correspondence, or a Hilbert–Pólya realization. Nor does an $o(k)$ counting remainder alone imply meromorphic continuation of a spectral zeta function.

All mathematical arguments needed for the stated theorems are given in the article. No numerical eigenvalue data, simulation, or experimental validation is used. The source manuscript is supplied as modular L^AT_EX with a four-entry bibliography; the build receipt records the initial source hashes, compiler invocation, and actual PDF and log hashes.

AI and review disclosure. AI assistants materially assisted the derivation, proof checking, source comparison, and preparation of this manuscript. The underlying proof package received a separate internal check before drafting. The assistant that performed that check also prepared this article; that earlier check is therefore not represented as an independent review of the present manuscript. No external referee approval or external-model assessment is claimed.

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